

Pravir Sinha

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PROFESSIONAL SNAPSHOT

AI Solutions Architect specializing in Agentic AI systems, RAG pipelines, and production multi-agent platforms across automotive, manufacturing, and supply chain domains. Hands-on builder of end-to-end AI architectures combining deep technical design with stakeholder advisory and enterprise deployment.

CORE SKILLS

Solution Architecture & AI Strategy	AI and ML Stack	Cloud AI Platforms	Domain
Agentic AI Systems, Service Intelligence Architecture, Enterprise Workflow Automation, HITL Design, 0-to-1 Delivery	LLMs, RAG, LangChain, LangGraph, PyTorch, FastAPI, Streamlit, PostgreSQL, Docker, CI/CD	AWS (SageMaker, Bedrock, S3, ECS, Lambda), GCP (Vertex AI, Cloud Run, GCS, BigQuery), Pinecone	Automotive, Supply Chain, Smart Manufacturing, CPG, FMCG, B2B SaaS, Aftersales & Field Service

PROFESSIONAL EXPERIENCE

AI Solutions Architect | Independent Practice Oct 2024 to Present

Production AI Systems

Conduit | Multi-Agent Automotive Service Management Platform

LangGraph | FastAPI | GPT-4o | Pinecone | PostgreSQL | Streamlit | GCP | Docker

- Architected a 5-agent autonomous service intelligence platform for automotive dealerships: intake, parts lookup, quoting, transaction, and replenishment agents handle the full workflow autonomously, turning a customer complaint into a priced repair order in under 45 seconds
 - Semantic parts search via vector DB; Human-in-the-Loop controls for high-value jobs and low-confidence classifications; deterministic safety guardrails override AI output on critical paths
- Outcomes: faster repair order turnaround, fewer parts stockouts, better customer experience at the counter
- Quality validated through evaluation layers (LLM-as-judge, rule-based checks) to ensure response accuracy and guard against hallucinations; LangSmith traces per-agent latency and cost per repair order

ShopMind | Real-Time Manufacturing Intelligence Platform

Python | FastAPI | LangChain | GPT-4o | PostgreSQL | GCP Cloud Run | Docker

- Plant managers received shift reports too slowly to act on machine problems before they escalated. Built four specialised agents that watch production lines continuously, detect early breakdown signals, identify compound machine crises, and let managers ask plain-language questions to understand what is going wrong and why
 - Blackboard architecture where agents share a common state layer rather than calling each other directly, keeping each agent independently testable and preventing cascading failures
- Outcomes: earlier warnings before breakdowns, reduced unplanned downtime, cleaner shift handover visibility
- AI response quality validated through BIRD-SQL eval methodology to track answer relevance, factual faithfulness, and hallucination detection; LangSmith observability for response latency and token cost per query

AI Training and Mentoring

- Delivered corporate workshops and one-on-one coaching sessions for more than 40 students and working professionals on LLM fundamentals, prompt engineering, RAG systems, LangGraph multi-agent orchestration, and Agentic AI for enterprise workflows
- Designed curriculum around practical use case design and hands-on building; participants ranged from engineering students to product and business professionals upskilling in applied AI

Manager, Enterprise Analytics and AI Solutions | **Accordion, Hyderabad** Jan 2022 to Sep 2024

- Demand Forecasting Platform | \$200M+ candy manufacturer

- Designed and deployed an LSTM-based batch forecasting pipeline on PostgreSQL feeding Power BI dashboards; led end-to-end solution design and supply chain integration; led 8-person cross-functional team across Data Engineering, ML Engineering, BI, and business stakeholders
- Reduced stockouts significantly; service levels improved from 82% to 94%, delivering 20% sales uplift
- Maintained model drift monitoring and performance dashboards for supply chain leadership post-launch
- **Automated Executive Reporting Platform with Text-to-SQL | \$500M+ snacks manufacturer**
 - Replaced fragmented Excel and manual BI reporting with an automated pipeline aggregating daily regional sales data into PostgreSQL, feeding a Power BI dashboard and a C-suite email snapshot delivered before 8am ET daily
 - Grounded an OpenAI-powered text-to-SQL layer on the PostgreSQL schema using prompt engineering and schema injection, enabling executives to query historical trends on demand
 - 60% reduction in manual FP&A effort; 95% C-suite adoption within 3 months; gathered requirements directly from executives and aligned roadmap with business priorities through regular reviews
- **RAG-Based Service Support Chatbot | Construction hoist manufacturer**
 - Built a RAG chatbot for supervisors managing safety-critical construction hoists, retrieving from equipment manuals and resolved ticket history to reduce reliance on after-sales calls for fault diagnosis; RAG architecture uses OpenAI embeddings with vector search over structured knowledge bases
 - Piloted with 60 supervisors across 7 construction companies over 6 weeks; reduced inbound calls to the after-sales team during the pilot

Strategy & Technology Consultant | Droege Group, Delhi *Apr 2019 to Oct 2020*

- Developed a 3-year product roadmap for a European contact centre headset manufacturer targeting \$80M to approximately \$110M in revenue; roadmap approved for \$15M investment
- Led market entry strategy for a US hydraulic technologies company in India; defined ICP, closed 5 distribution partnerships, and delivered \$2.5M in Year 1 revenue

Strategy & Analytics Consultant | Boston Analytics | Tecnova India Pvt. Ltd. *Apr 2016 to Apr 2019*

- Delivered growth strategies, market entry analysis, and supply chain optimizations for international clients across automation, hoisting, and industrial fasteners
- Cut manufacturing lead times from 5 weeks to under 1 week; grew market share from 18% to 35% for a European hoist manufacturer

Assistant Manager, Quality Control | Maruti Suzuki India Ltd., Gurgaon *Jun 2011 to Feb 2015*

- Managed quality control for gear and shaft machining lines in high-volume automotive manufacturing; built deep OEM and supply chain domain grounding in enterprise service workflows, fault resolution, and cross-functional escalation
- Led cross-functional defect resolution across customers, suppliers, and production teams; approximately 15% YoY defect reduction

EDUCATION

B.Tech, Industrial Engineering | *NIT Jalandhar* | *2007 to 2011* | *GPA: 7.38 / 10*

CERTIFICATIONS

- AWS Certified AI Practitioner (2025)
- AI Product Manager Nanodegree, Udacity (2025)